

Lithium Acetate Transformation of *Saccharomyces cerevisiae*

This protocol describes the rapid transformation of *Saccharomyces cerevisiae* using the lithium acetate/single-stranded carrier DNA/polyethylene glycol (LiAc/SS-DNA/PEG) method. The goal is to introduce plasmid DNA into yeast cells and recover transformants on selective medium. In this method, yeast cells grown on a YPAD agar plate are mixed with lithium acetate, PEG 3350, boiled salmon sperm carrier DNA, and plasmid DNA, then incubated at 42°C before plating. This rapid protocol is suitable when only a small number of transformants are required.

**Day 1 → Thursday

**Day 2 → Friday (incubate over weekend)

Reagents

✓	Name	Quantity	Location/Notes
☐	<i>S. cerevisiae</i> [BY4741] strain	1 culture/YPAD patch	Yeast strain to be transformed, retrieved from agar
☐	YPD	3mL (Day 1) ~5mL (Day 2)	Overnight growth
☐	Sterile water	As needed	Resuspension and plating dilutions
☐	Lithium acetate, 1.0 M (3.3g LiOAc + ddH ₂ O to 50mL, filter sterilize)	15 µL per transformation	Transformation mix (T-mix) reagent
☐	Lithium acetate, 0.1 M	300 µL	Resuspending yeast cells
☐	PEG 3350, 50% (w/v)	120 µL per transformation	Transformation mix (T-mix) reagent

☐	ss carrier DNA (salmon sperm, 10 mg/mL)	5 μ L per transformation	Boil for 5 mins, chill before use
☐	Plasmid DNA	0.5 μ g per transformation	Add in 25 μ L total with water
☐	SC Selective agar plates	1 plates per transformation	Marker to plasmid match

Equipment

✓	Name	Quantity	Location/Notes
☐	1.5 mL microcentrifuge tubes	Several	Transformation reaction
☐	Micropipettes and sterile tips	1 set	Measuring and transferring reagents
☐	Microcentrifuge	1	13,000 rpm
☐	Vortex mixer	1	Resuspension and mixing of carrier DNA
☐	42°C water bath or heat block	1	Transformation incubation
☐	Ice bucket or ice-water bath	1	Chilling carrier DNA
☐	30°C incubator	1	Growing yeast and incubating plates
☐	Sterile spreaders or plating beads	As needed	Plating transformants
☐	Markers and labels	As needed	Tubes and plates

Instructions

1. Prepare yeast cells (Day 1)

- a. Inoculate 3mL of YPD with a colony of *S. cerevisiae* BY4741
- b. Incubate **overnight at 30°C**, 250 RPM

2. Harvest yeast cells

- a. Label a 15 mL tube for the transformation reaction.
- b. Dilute a new culture 1:50 (overnight culture: YPD), this should read approximately 0.2 on OD600 with a total volume of 5mL
- c. Grow to OD600~ 0.6-1.0 (4-5h at 30C, 250rpm)
- d. Pellet the cells by centrifuging at **3000-4000 rpm for 1-5 min** at room temperature.
- e. Discard the supernatant **carefully**.
Note: Be careful not to lose the pellet when removing the supernatant, as this contains the yeast cells needed for transformation.
- f. Resuspend the cells in 300 µL of 0.1M Lithium acetate
Note: Make sure the pellet is fully resuspended, since incomplete mixing can reduce transformation efficiency.

Reagent Tracker (use this table as needed)	Quantity
☐ Overnight culture	100 µL
☐ YPD	4900 µL
☐ 0.1 Lithium acetate	300 µL

3. Prepare carrier DNA (Day 2)

- a. Label a 1.5 mL microcentrifuge tube for the carrier DNA.
- b. Heat one tube of **single-stranded carrier DNA** (salmon sperm DNA, 10 mg/mL) to **100°C** for **2 minutes** (use PCR machine).
- c. **Immediately** chill the tube in ice/water.
- d. Vortex briefly
Note: Carrier DNA must be boiled and then quickly chilled before use. This helps denature the DNA and improves transformation efficiency.
Note: Vortex the carrier DNA before pipetting to make sure it is evenly mixed.

Reagent Tracker (use this table as needed)	Quantity
☐ ss carrier DNA (salmon sperm, 10 mg/mL)	5 µL EACH

4. Prepare the transformation mix (T-mix)
- a. Create a fresh solution of 40% PEG 0.1M LiOAc (the following is the amount for ONE transformation (150µL but 140µL is used), scale accordingly):

Reagent Tracker (use this table as needed)	Quantity
☐ PEG 3350, 50% (w/v)	120 µL
☐ Lithium acetate, 1.0 M	15 µL
☐ ddH ₂ O	15 µL

- b. Add the following to a 1mL tube **IN THIS ORDER**:

Reagent Tracker (use this table as needed)	Quantity
☐ Plasmid DNA (+ sterile water)	0.5 ug / 25 µL
☐ ss carrier DNA (salmon sperm, 10 mg/mL)	5 µL
☐ Yeast cells	30 µL
☐ 40% PEG 0.1M LiOAc	140 µL

Note: PEG is very viscous, so pipette slowly and carefully to ensure the correct volume is added.

- c. Use **0.5 µg plasmid DNA** within the total 25 µL plasmid DNA + sterile water volume.
- i. The choice of 25 µL is arbitrary but it is important that the plasmid mass NEEDS to be at least 0.5 µg to ensure effective transformation

- d. Invert the tube multiple times to ensure the suspension is mixed properly
 - e. Ensure the tube is labelled before incubation.
5. Incubate transformation reaction
 - a. Incubate the mix at room temperature for 30 minutes
 - b. Heat shock the transformation tube at **42°C for 20 minutes**.
6. Recover transformed cells
 - a. Centrifuge the tube at **3,000 rpm for 1-2 min** at room temperature.
 - b. Remove the transformation mix carefully with a pipette.
 - c. Add **1.0 mL sterile water**.
 - d. Resuspend the cells thoroughly by breaking up with pipette

Note: When removing the transformation mix, avoid disturbing the pellet.

Note: Resuspend the cells completely before plating to ensure even colony distribution.

Reagent Tracker (use this table as needed)	Quantity
☐ Sterile water	1.0 mL

7. Plate transformed cells
 - a. Label the SC selective agar plates with the strain name, plasmid name, medium type, date, and plating volume.
 - b. Plate **100 µL** of the resuspended cells onto the appropriate SC selective agar plates.
 - c. Spread the samples evenly using sterile spreaders or plating beads.
 - d. Incubate plates at **30°C for 2-3 days, or until visible isolated colonies appear**.
 - e. Isolate transformant colonies from selective plates.

Note: Spread the cells evenly to avoid colony crowding.

Footnotes

The LiAc/SS-carriers DNA/PEG method includes four protocol variations: rapid, high-efficiency, large-scale, and microtiter plate. The rapid protocol is intended for introducing a plasmid into a yeast strain when only a small number of transformants is needed. The high-efficiency protocol is used when large numbers of transformants are required or when transforming, integrating plasmids, DNA fragments, or oligonucleotides. The large-scale protocol is mainly for screening complex plasmid libraries, and the microtiter plate protocol is designed for high-throughput transformation of multiple strains or samples. For this project, the rapid protocol was selected because the goal was routine plasmid introduction into *S. cerevisiae* for a student-team proof-of-concept, rather than large-scale screening or recovery of high transformant numbers.

References

Gietz, R Daniel, and Robin A Woods. "Transformation of yeast by lithium acetate/single-stranded carrier DNA/polyethylene glycol method." *Methods in enzymology* vol. 350 (2002): 87-96. doi:[10.1016/s0076-6879\(02\)50957-5](https://doi.org/10.1016/s0076-6879(02)50957-5)

S. cerevisiae episomal plasmid quick transformation (**Dr. Klemm**)